

Risk-Aware Distributional RL for 5G-Advanced Handover Decisions

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1. The Problem

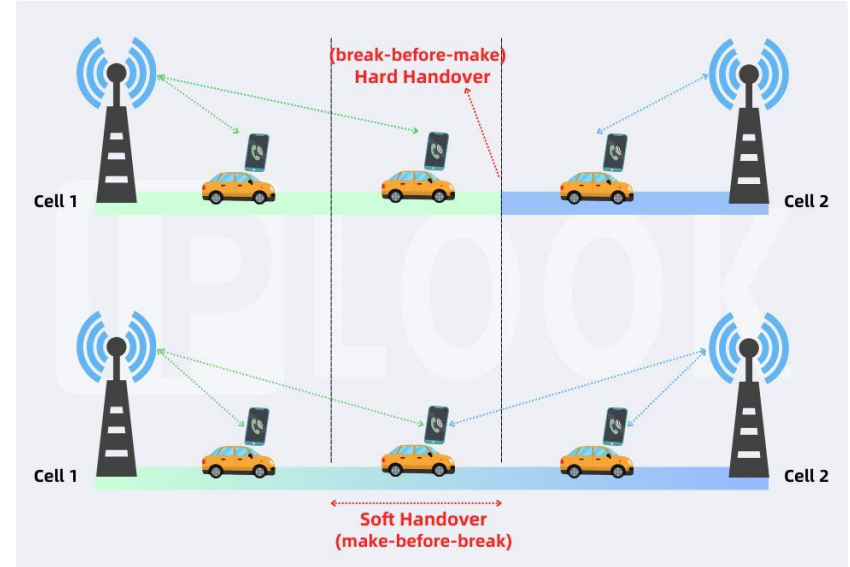
What Is a Handover?

A moving UE (User Equipment) is served by one cell/sector at a time.

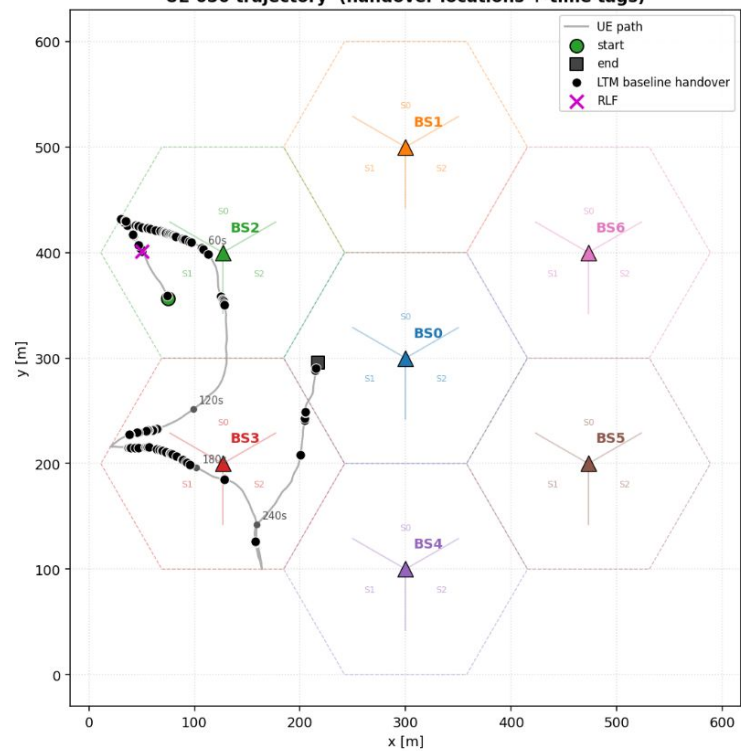
Serving link weakens and a neighbor grows stronger

⇒ the network must switch serving cell!

⇒ a **Handover (HO)** takes place

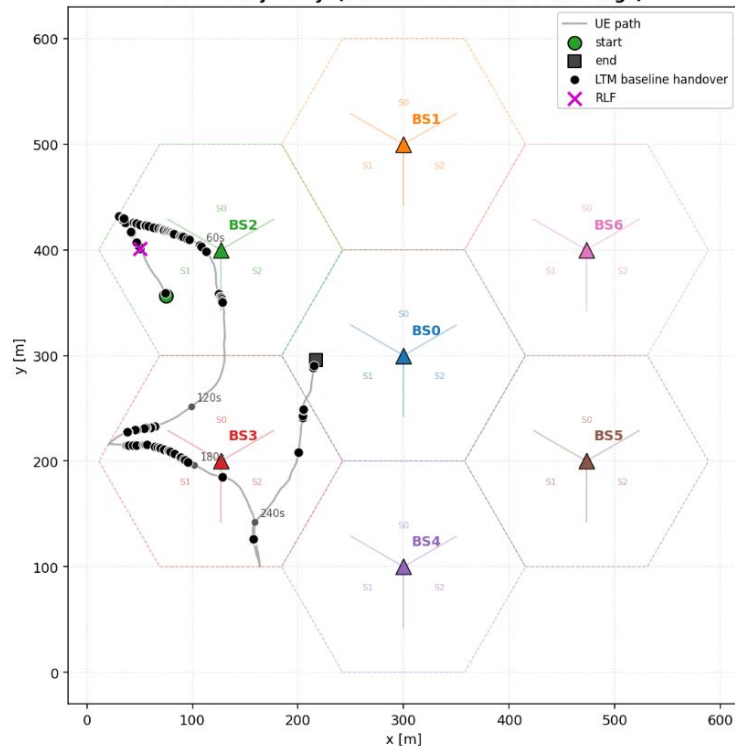


UE 650 trajectory (handover locations + time tags)



LTM baseline — UE 650: trajectory, serving quality, and cell decisions

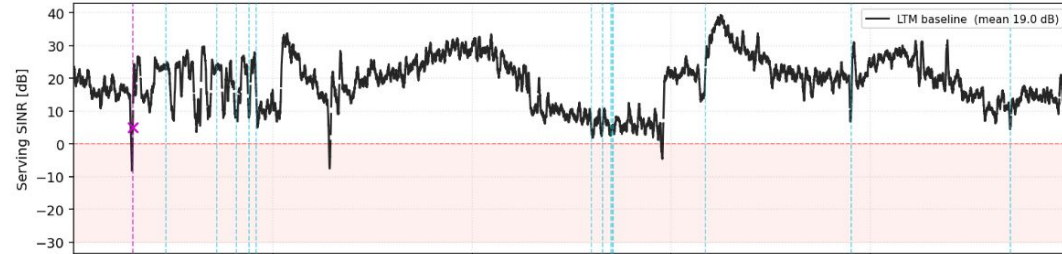
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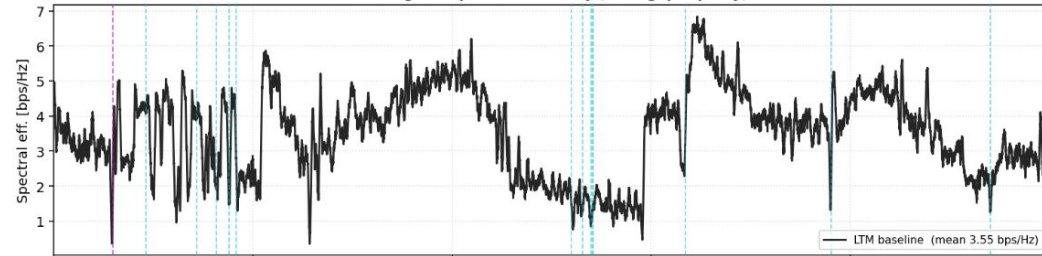
Episode scoreboard

KPI (per UE)	LTM
Handovers / ep	102
Ping-pongs	34
HO failures (HOF)	13
Radio-link fails (RLF)	1
Mean SINR [dB]	18.97
Mean SE [bps/Hz]	3.55
Avg prepared cells	1.49

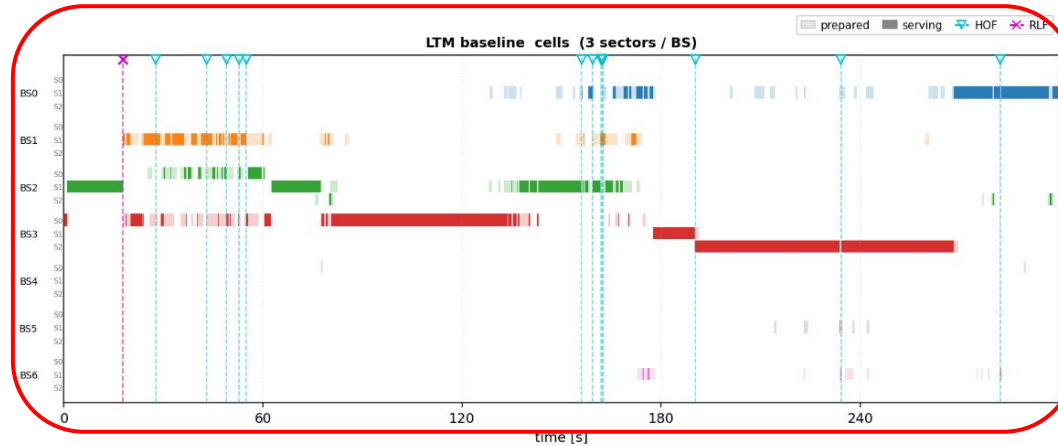
Serving-cell SINR (cyan ∇ = HOF, magenta $\times$ = RLF, shaded = OOS zone)



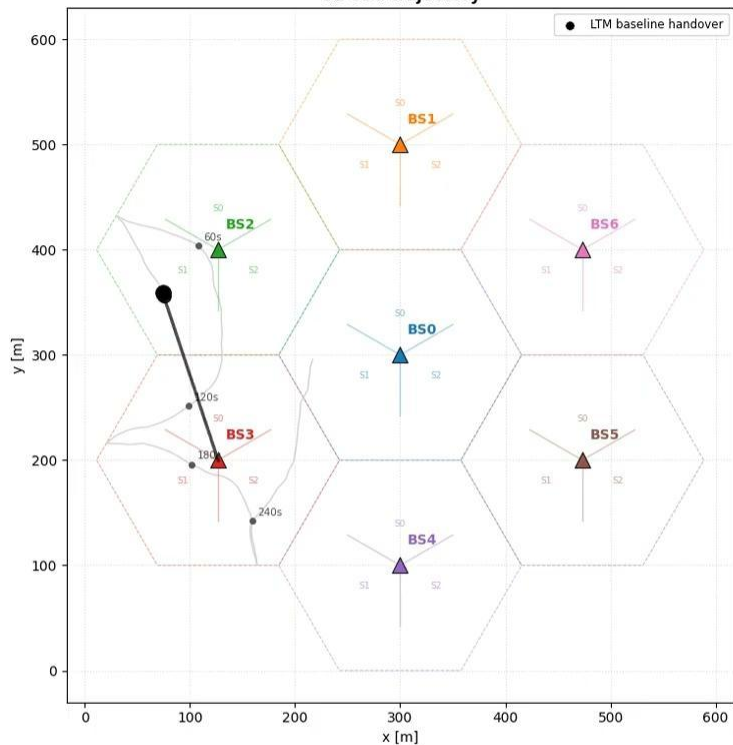
Serving-cell spectral efficiency (throughput proxy)



LTM baseline cells (3 sectors / BS)



UE 650 trajectory



Episode scoreboard

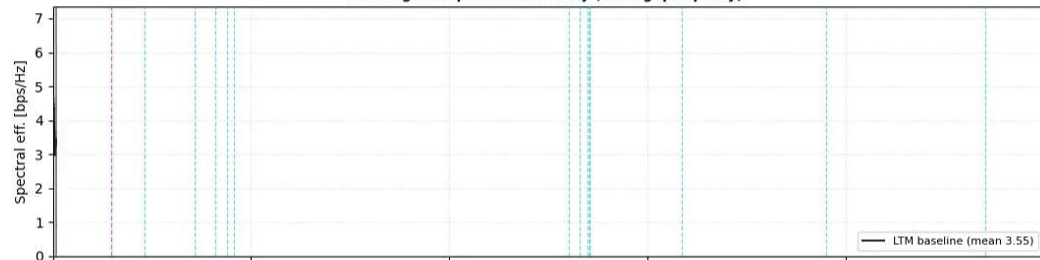
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LTM baseline — UE 650

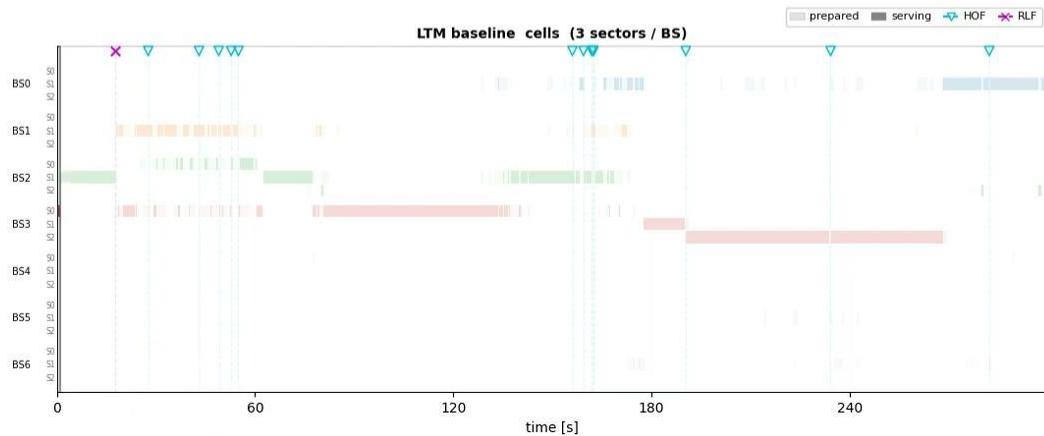
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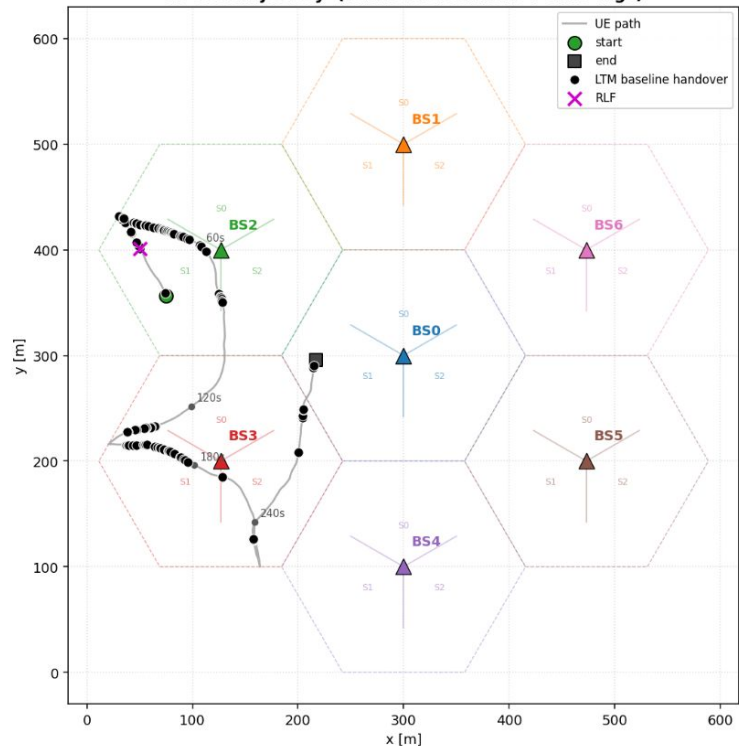
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UE 650 trajectory (handover locations + time tags)



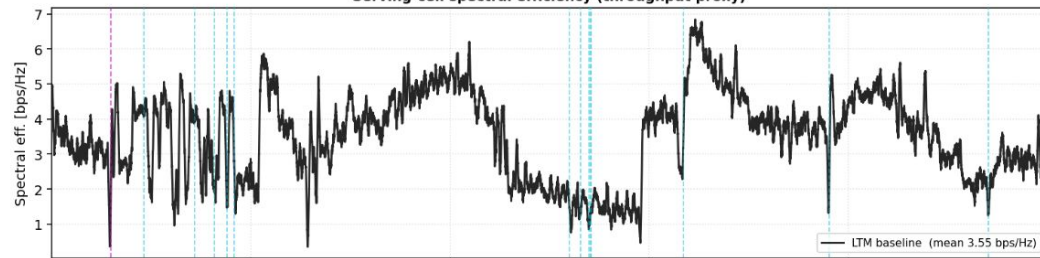
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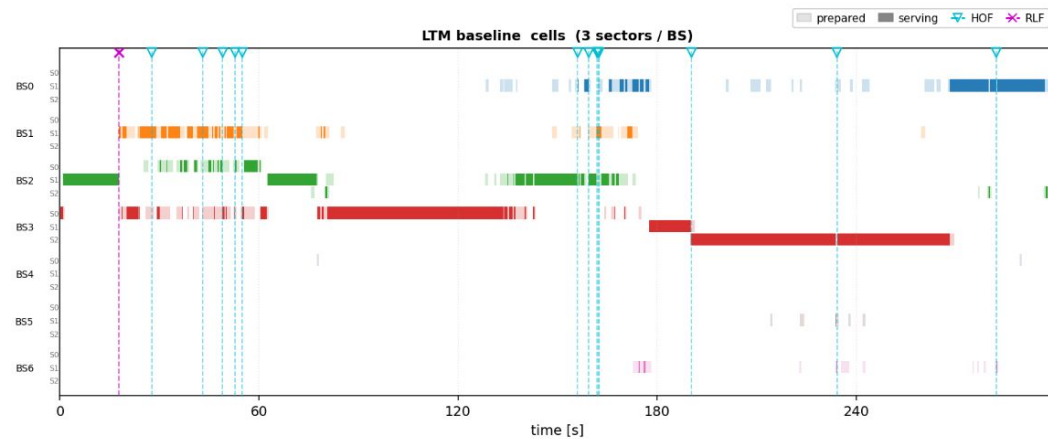
Serving-cell SINR (cyan ∇ = HOF, magenta $\times$ = RLF, shaded = OOS zone)



Serving-cell spectral efficiency (throughput proxy)

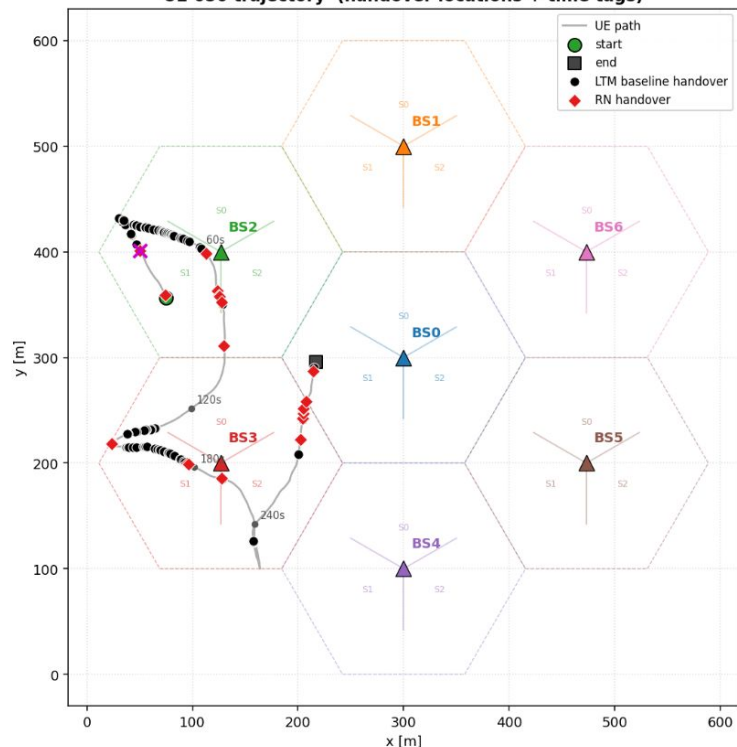


LTM baseline cells (3 sectors / BS)



LTM baseline vs. RN — same UE 650, different handover decisions

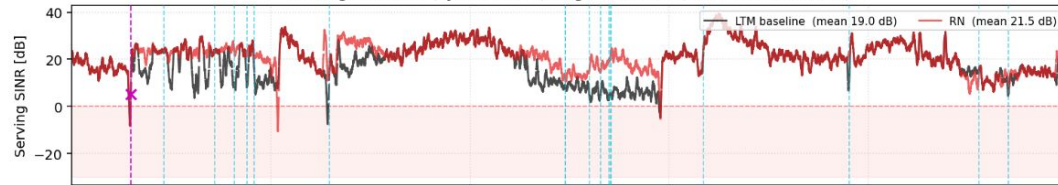
UE 650 trajectory (handover locations + time tags)



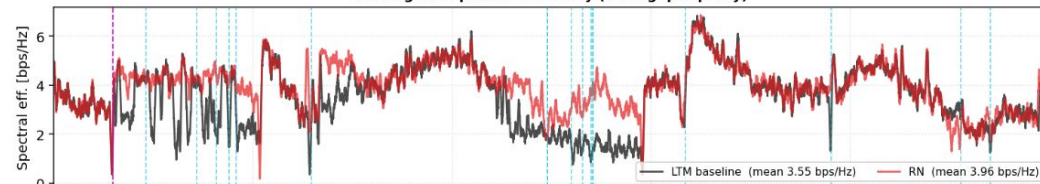
Episode scoreboard (green = better)

KPI (per UE)	LTM	RN
Handovers / ep	102	23
Ping-pongs	34	2
HO failures (HOF)	13	4
Radio-link fails (RLF)	1	1
Mean SINR [dB]	18.97	21.49
Mean SE [bps/Hz]	3.55	3.96
Avg prepared cells	1.49	2.13

Serving-cell SINR (cyan ▽ = HOF, magenta X = RLF, shaded = OOS zone)



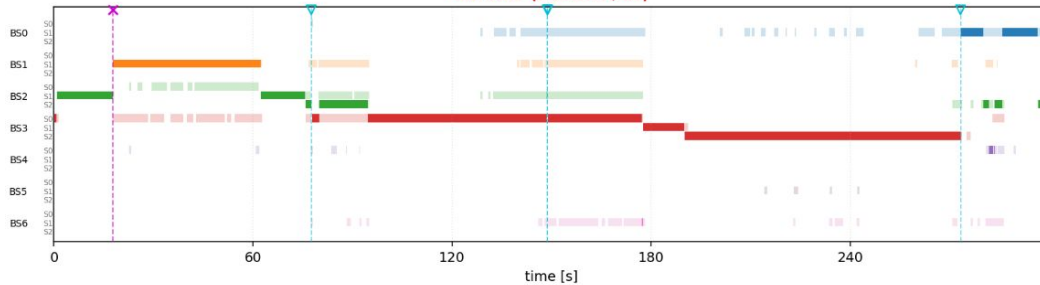
Serving-cell spectral efficiency (throughput proxy)



LTM baseline cells (3 sectors / BS)



RN cells (3 sectors / BS)



Objectives:

A good handover policy should:

- Maximize quality: high **Capacity** (bps/Hz) and **Reliability** (%)
- Minimize churn: as few **Handovers** and **Ping-Pongs** as possible
- Avoid faults: **Handover Failures** and **Radio Link Failures** are rare but catastrophic

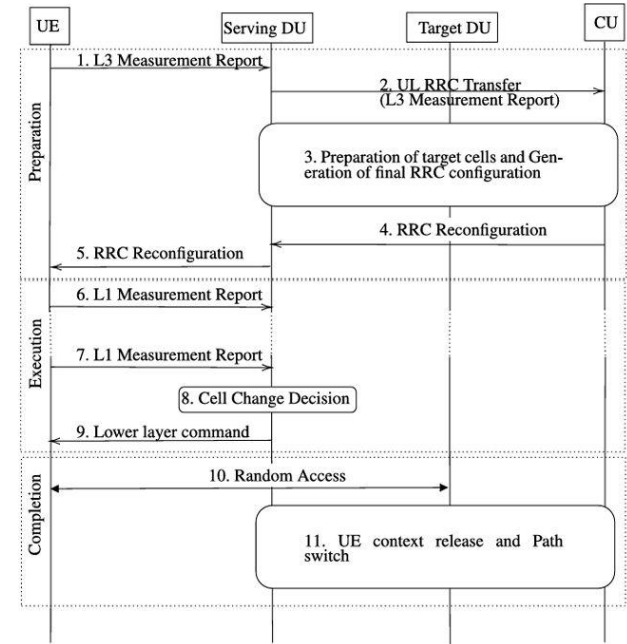
Handing over more aggressively buys quality but invites faults and churn.
The policy must balance them.

2. Baselines & Prior Art

LTM: Lower-Layer Triggered Mobility (L1/L2)

1. Prepare a neighbor early

when RSRP rises to within 3 dB of the serving cell
(pre-establish its RRC config so the switch is fast)



(b) High-level signalling diagram of lower layer mobility.

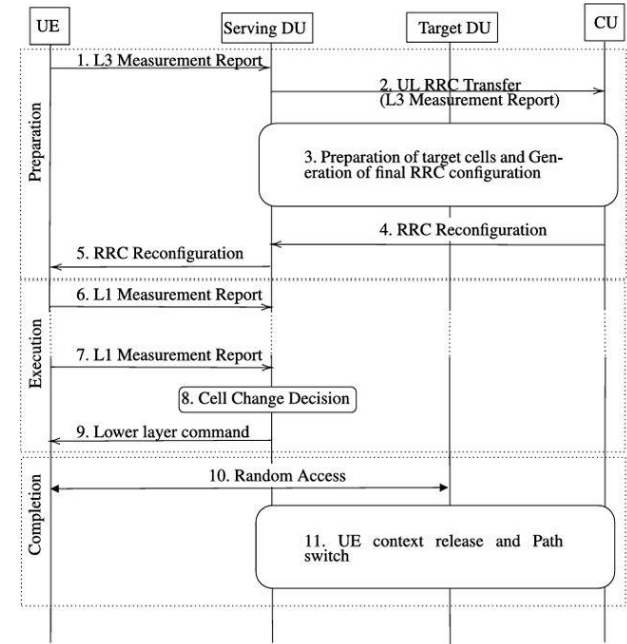
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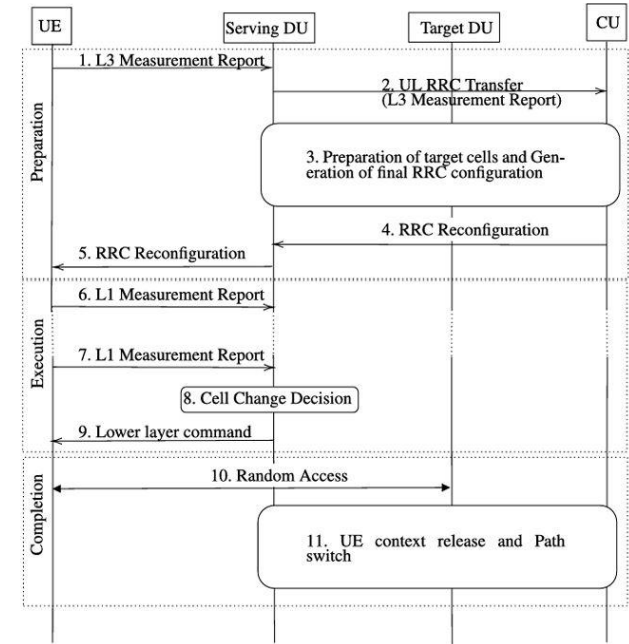
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2. Execute the handover greedily

when a neighbor's RSRP exceeds the serving by 3 dB

Up to 5 cells prepared at once.

But preparing isn't free: costs signalling (Preparations)
and reserves radio resources (Resource Reservation).



(b) High-level signalling diagram of lower layer mobility.

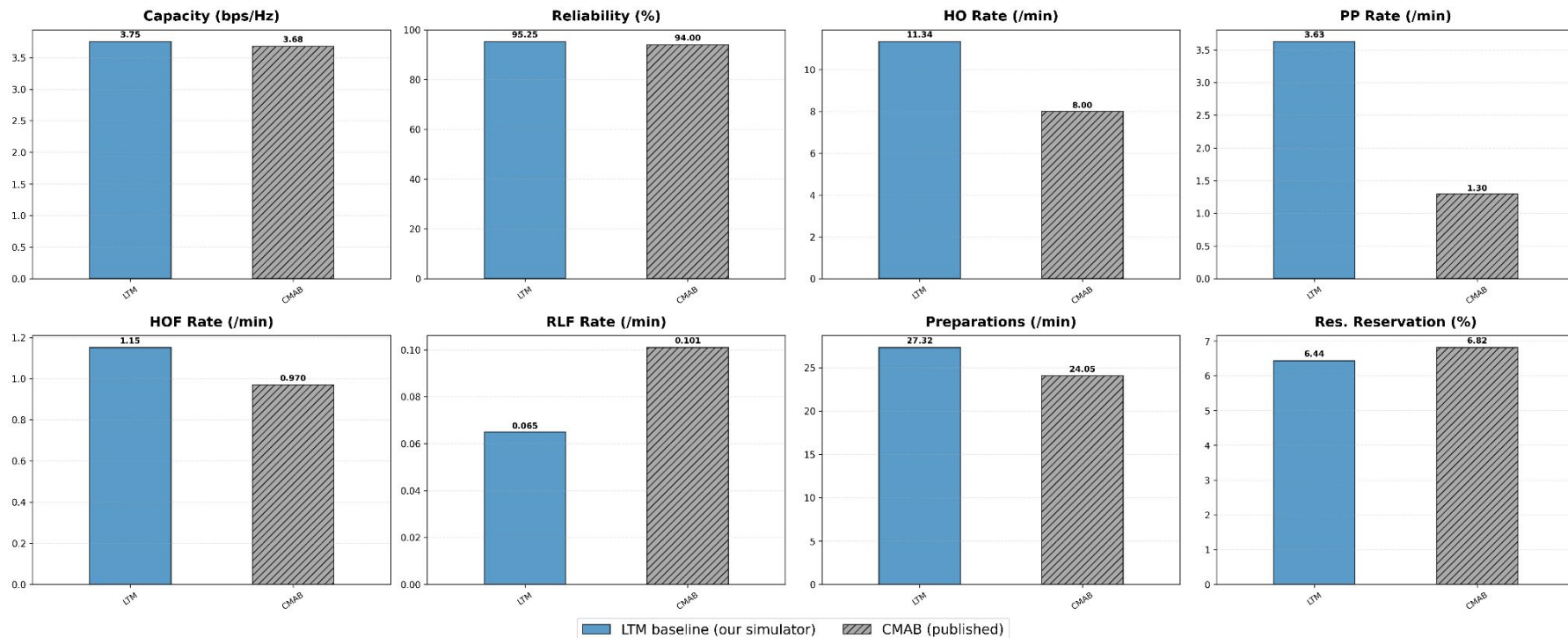
Prior art: Contextual Multi-Armed Bandits [1]

First learning-based approach to LTM handover — picks among prepared candidates from context. Uses UCB + custom kernel.

- Beats LTM on churn (fewer Handovers / Ping-Pongs).

Prior art: Contextual Multi-Armed Bandits [1]

Prior-Art Baselines: LTM vs CMAB



Promising results — but each decision changes the next: the problem is sequential.

3. A Markov Decision Process

The State - an 88-dim vector

Context:

- UE **speed**
- UE **position** (x, y)
- serving-cell **tenure** — time on the current cell since last HO

Serving identity:

- one-hot serving sector (×21)

Per-sector radio measurements (×21 each):

- **RSRP** — received reference power (interference-blind → coverage / LTM trigger)
- moving-avg **SINR** — link quality (signal vs interference + noise)
- moving-avg **MCS** — spectral efficiency = Capacity

The Actions

- Action = **which sector serves the UE next** $\Rightarrow$ stay, or switch to another sector.
- $7 \text{ BS} \times 3 \text{ sectors} = 21$ candidate sectors.
- **Masked** to the feasible set each step: $\{\text{serving}\} \cup \{\text{prepared}\}$
The agent can only stay or hand-over to an already prepared cell (≤ 5).

The Reward

Base: throughput R_{thr} (the spectral efficiency you got this step).

Multiplied by:

- a penalty each time an event fires: Handover $\times 0.8$, Ping-Pong $\times 0.9$, Handover Failure $\times 0.1$
- a reliability factor (inverse sigmoid) that shrinks as the link goes out-of-sync, toward RLF

$$R_t = R_t^{\text{thr}} \cdot \frac{\alpha_{\text{HO}}^{\mathbb{I}_t^{\text{HO}}} \alpha_{\text{PP}}^{\mathbb{I}_t^{\text{PP}}} \alpha_{\text{HOF}}^{\mathbb{I}_t^{\text{HOF}}}}{1 + e^{2(N_t^{\text{OOS}} - 2)}}$$

A Note on the Reward

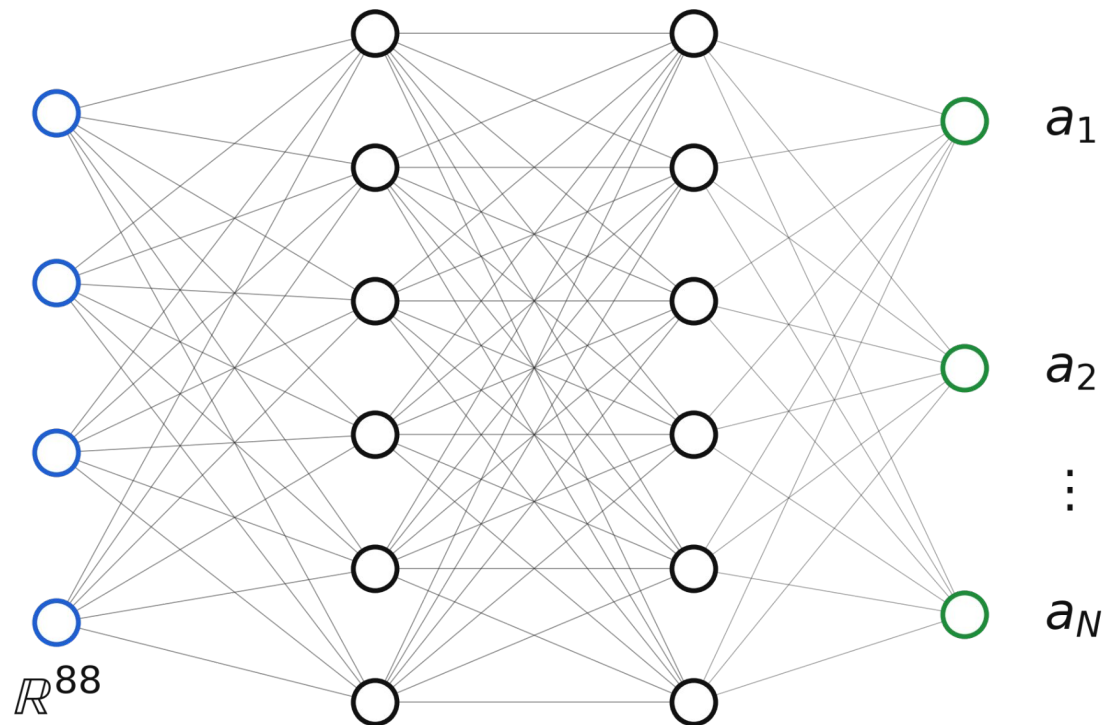
- Actions/decisions are **event-driven**: they fire on triggers, not on a fixed clock. So different policies make different numbers of decisions per episode.
- Maximizing total reward then rewards acting more often, not deciding better!
- So after training, we compare policies by reward per decision:
reward per decision = total reward $\div$ number of decisions

Solving the MDP: DQN

We learn the value of each state-action pair

- DQN: a network maps state $\rightarrow Q(s, a)$ = expected discounted return; act by $\operatorname{argmax}_a Q(s, a)$.
- ...but $Q(s, a)$ is only the mean — it collapses the whole distribution of possible returns into a single number.

Deep Q-Network (DQN)



What if we try to predict the whole distribution instead of the mean value?

4. Distributional Reinforcement Learning

The distributional Bellman Equation

Model the whole return distribution $Z(x,a)$, not just its mean Q .

$$Z(x, a) \stackrel{D}{=} R(x, a) + \gamma Z(X', A')$$

Learning the distribution gives a richer training signal!

The distributional Bellman Equation

$$Z(x, a) \stackrel{D}{=} R(x, a) + \gamma Z(X', A')$$

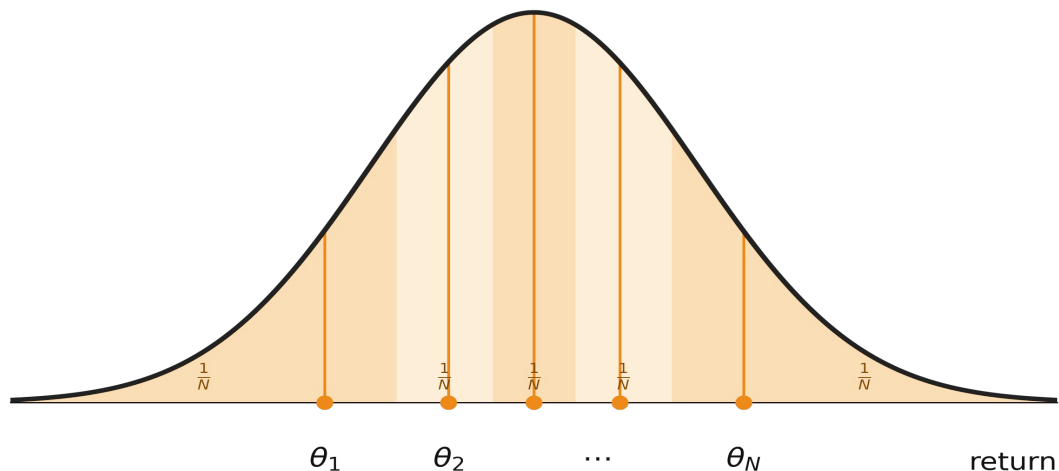
Q1: How do we represent $Z(x, a)$?

Q2: How can we “learn” the distribution?

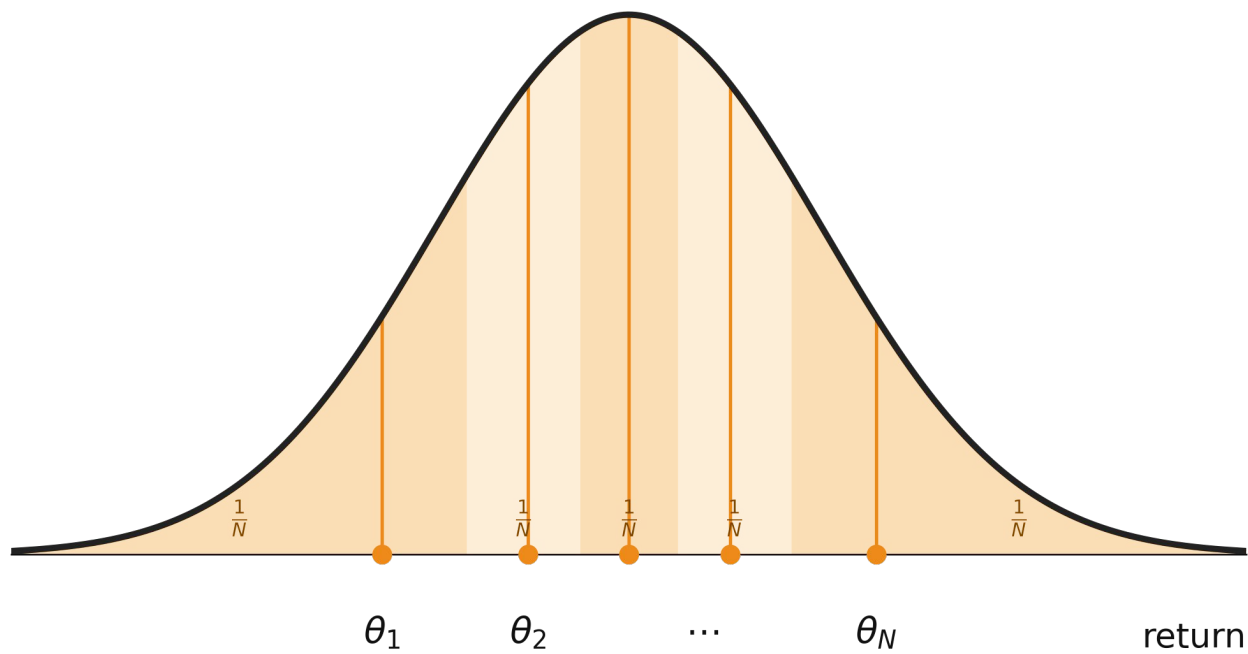
Q3. How do we act on it? Which action is best?

Q1: How do we represent Z? $\rightarrow$ with N quantiles

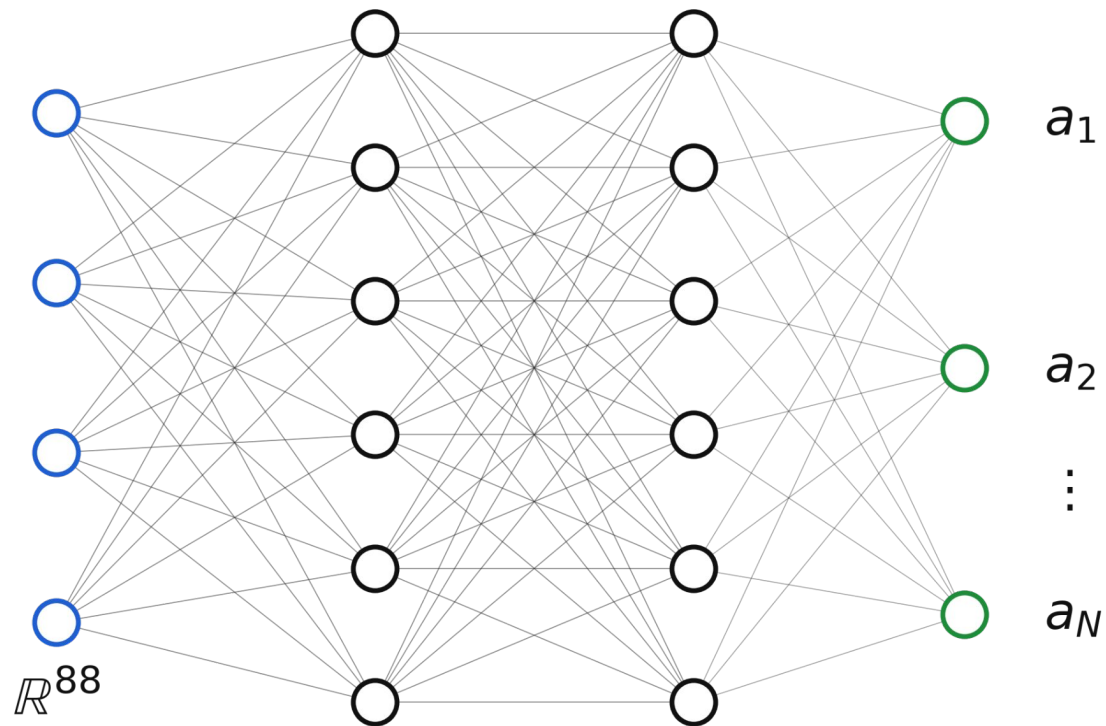
- The i-th output is the return below which a fraction τ_i of outcomes fall:
 $\theta_i \approx F^{-1}(\tau_i)$ (inverse CDF).
- Fix $\tau_i = (i-1/2) / N$ evenly in $[0,1] \rightarrow$ **N equal-probability slices**, the head learns the value of each θ_i based on the context.



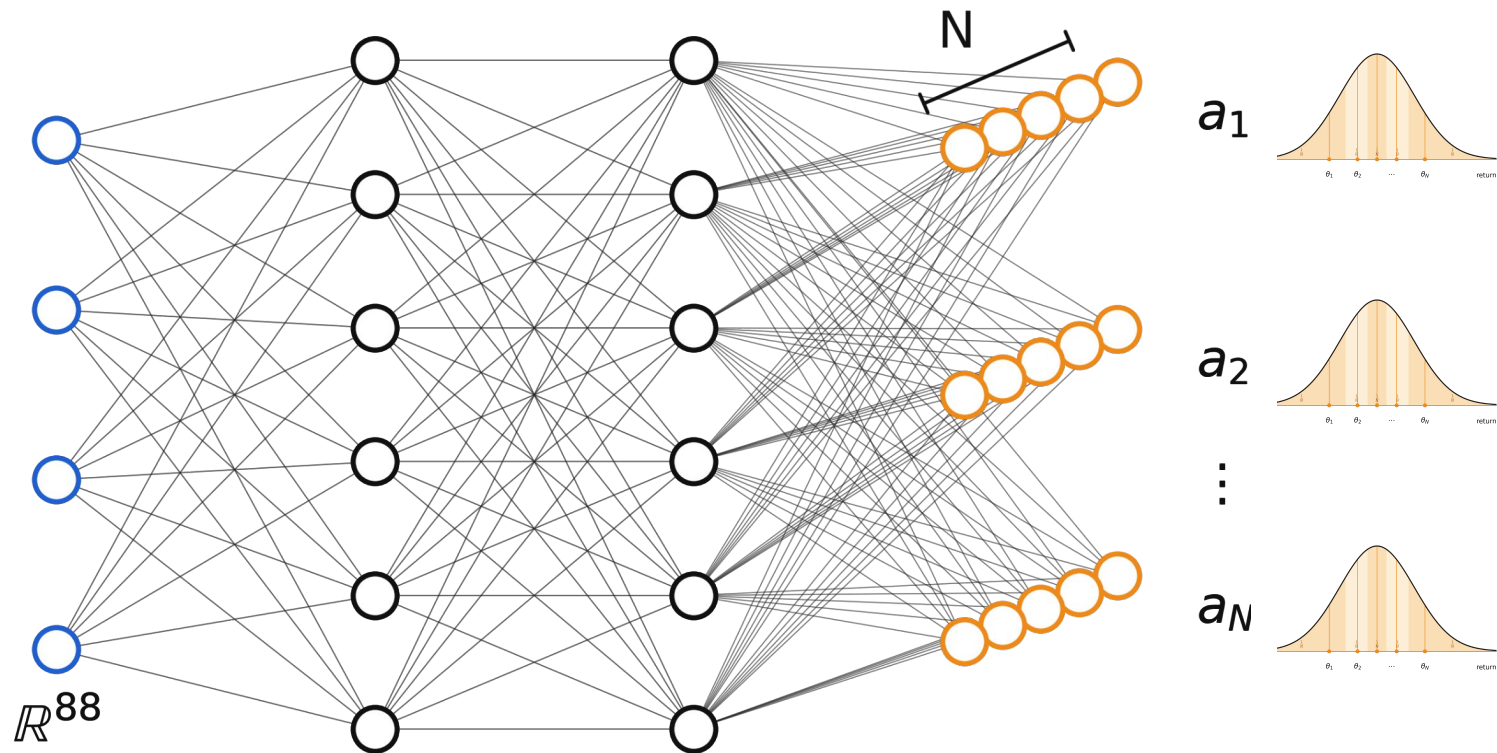
Q1: Representing Z by N quantiles



DQN



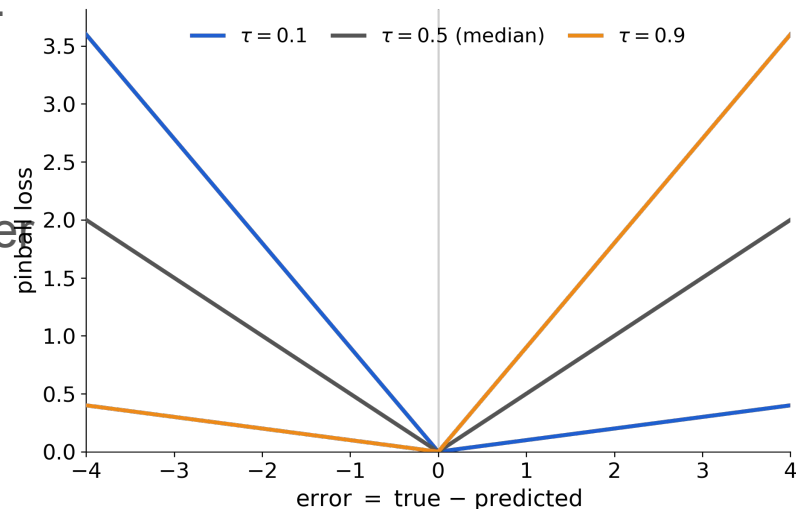
QR-DQN



Q2: Learning the distribution — The Pinball loss

An asymmetric absolute-error loss; the tilt is set by τ — slope τ above the prediction, $1-\tau$ below.

- A low quantile is punished hard when the true return lands below it, gently when above $\rightarrow$ it gets pushed down into the lower tail
- Minimizing it drives each output θ_i to its τ_i -quantile of the target (instead of the mean).

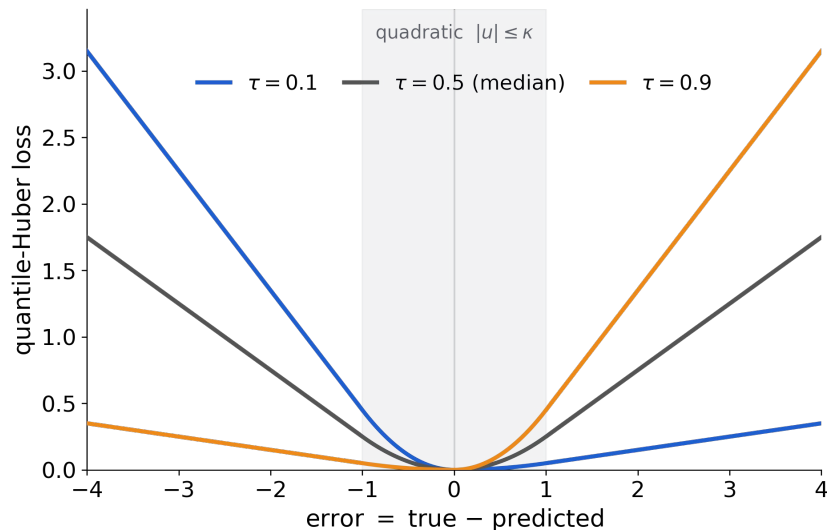


Q2: Huberize for smooth gradients

Round the sharp kink at 0 into a quadratic on $|u| \leq \kappa \rightarrow$ Opt. at $\kappa = 0.25$ in our runs.

$$\mathcal{H}_{\kappa}(u) = \begin{cases} \frac{1}{2}u^2, & |u| \leq \kappa \\ \kappa(|u| - \frac{1}{2}\kappa), & |u| > \kappa \end{cases}$$

$$\rho_{\tau}^{\kappa}(u) = |\tau - \mathbb{I}\{u < 0\}| \mathcal{H}_{\kappa}(u)$$



The loss for one quantile θ_i

- Target = a TD bootstrap, one per quantile:
 $y_j = r + \gamma \theta_j(x', a^*)$
- The N targets $\{y_j\}$ are N equally-likely representative values of the next-state return $Y = r + \gamma Z(X', A')$.
- We want θ_i to be the τ_i -quantile of that target return Y — so we average its pinball loss against all N targets.

$$\sum_{j=1}^N \rho_{\tau_i}^{\kappa}(y_j - \theta_i(x, a))$$

$$y_j = r + \gamma \theta_j(x', a^*)$$

Minimizing this pulls θ_i to the τ_i -quantile of the bootstrapped return.

Full loss function

Sum the single-quantile loss over all N quantiles and average. Each θ_i is regressed against the whole target distribution $\{y_j\}$ — distribution-to-distribution.

$$\mathcal{L}(x, a) = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N \rho_{\tau_i}^{\kappa}(y_j - \theta_i(x, a))$$

$$y_j = r + \gamma \theta_j(x', a^*)$$

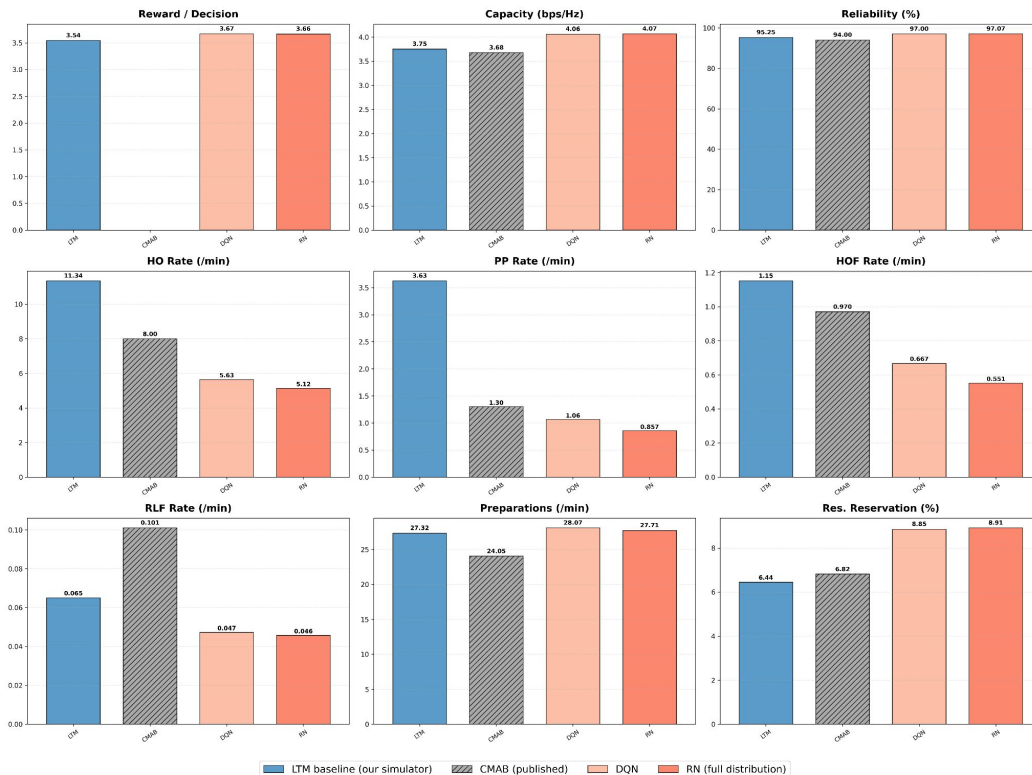
Q3: What to act on?

The N quantiles describe the whole return — but to pick an action we must collapse them to one number.

- Risk-neutral: act on the mean, $E[Z] \approx (1/N) \sum \theta_i$, then argmax. → RN agent.
- *But the mean is just one way to read the distribution...* (next section)

The distribution alone already helps — at equal reward

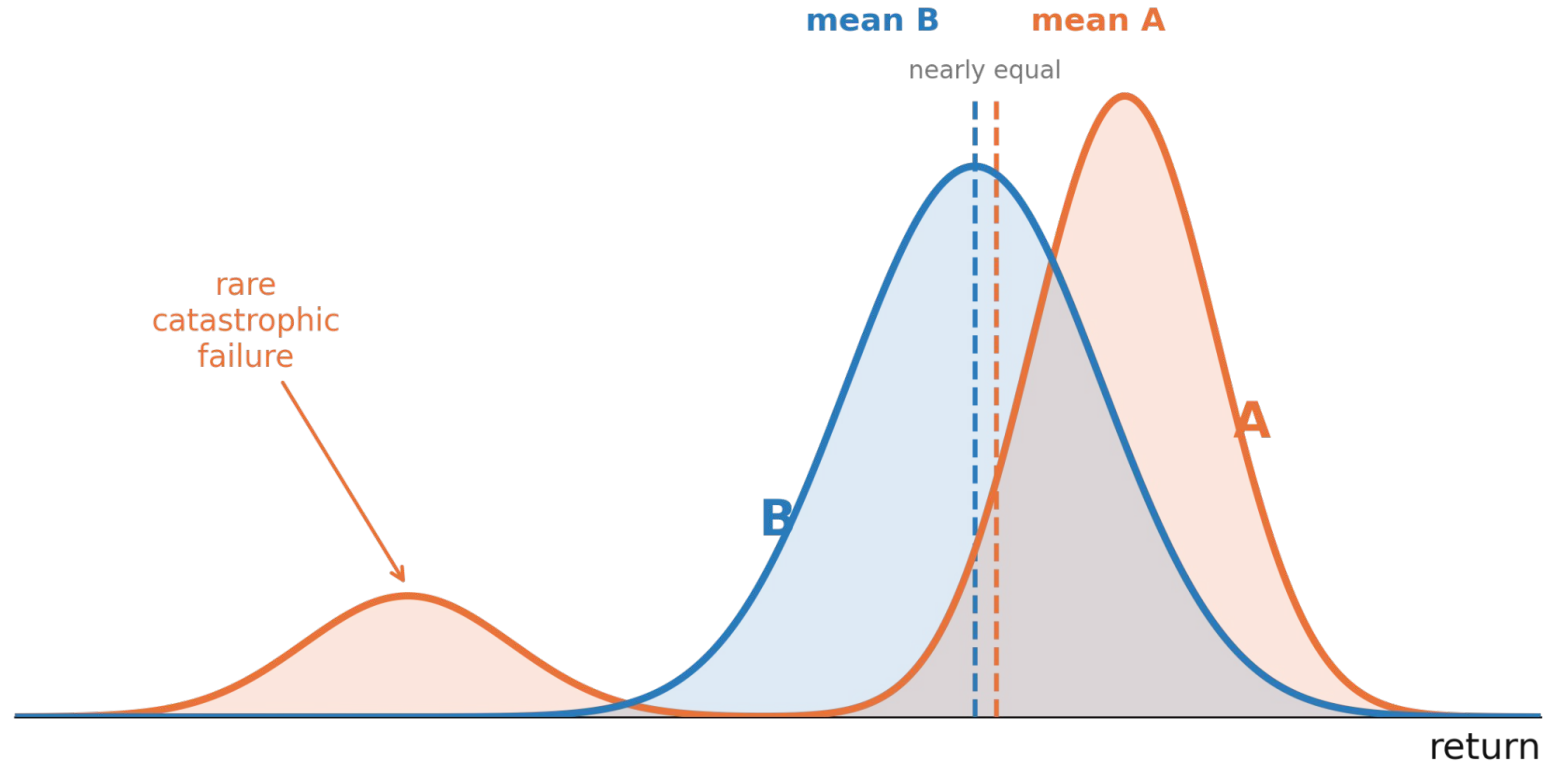
Distributional Learning on the Handover Task: DQN vs QR-DQN



Can we do better than acting on just the mean of the distribution?

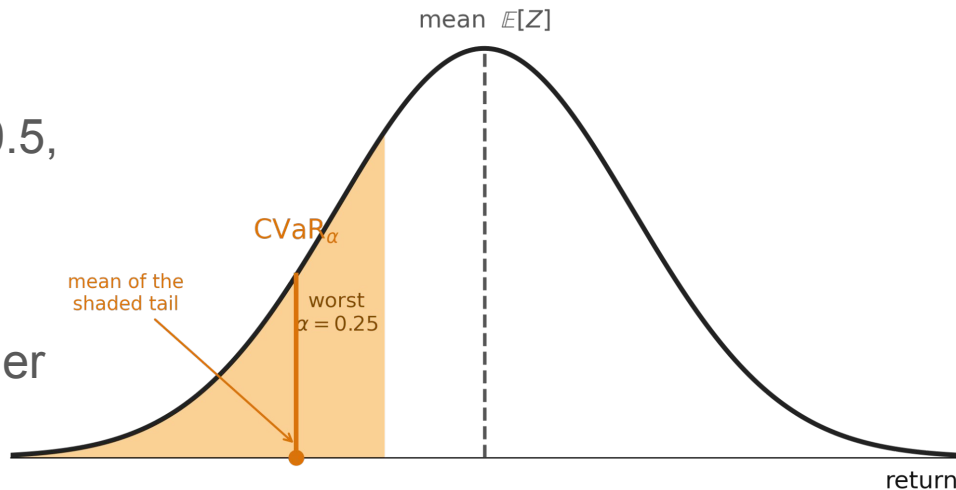
5. Risk-Aware Distributional RL

Which action should the model take?



CVaR: Conditional Value at Risk

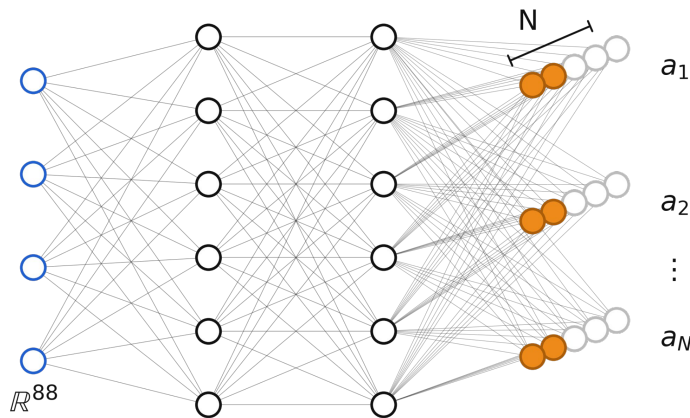
- CVaR_α = the average of the worst α -fraction of returns (the lower tail).
- e.g. uniform on $[-1, 1]$: $\text{CVaR}_{0.5} = -0.5$, the mean of the worse half.
- For handover: avoiding worst-case HOF/RLF is worth more than a higher average.



CVaR on the Quantile Head: average the lowest k

We already predict N quantiles $\rightarrow$ CVaR = average the lowest $k = \lceil \alpha N \rceil$ of them.

- Same network, same training — we only change the aggregation (the Q3 knob).
- **Re-weighting:** the full distribution is still learned; risk lives in the weights.



The risk dial α

- $\alpha=1 \rightarrow$ the mean (RN); smaller $\alpha \rightarrow$ deeper into the tail.
- Monotone trade: less $\alpha \rightarrow$ fewer failures, slightly less reward.
No interior optimum — just a controllable exchange.
- Knee at $\alpha = 0.25$
(α : $0.5 \rightarrow 0.25 \rightarrow 0.1$: HOF $0.44 \rightarrow 0.37 \rightarrow 0.35$; reward barely moves)

Soft-step: a graded dial

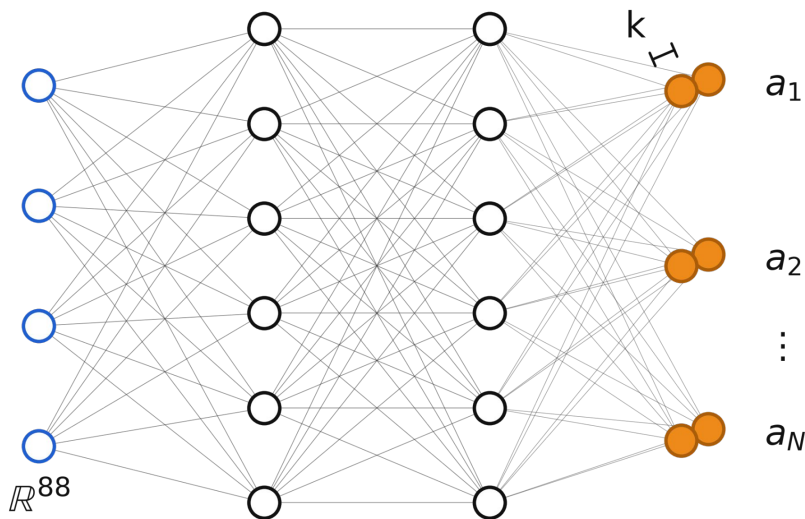
- Hard CVaR is a 0/1 step. Soft-step gives the bottom k quantities $\times r$ weight:
 $r=1 \Rightarrow \text{mean (RN)}$, $r \rightarrow \infty \Rightarrow \text{hard CVaR}$.
- Adopt $r = 2$ — most of the gain at the cheap end of the frontier.

Do we really need the full distribution to approximate the CVaR?

Every method so far keeps all N quantiles but acts on only the bottom k

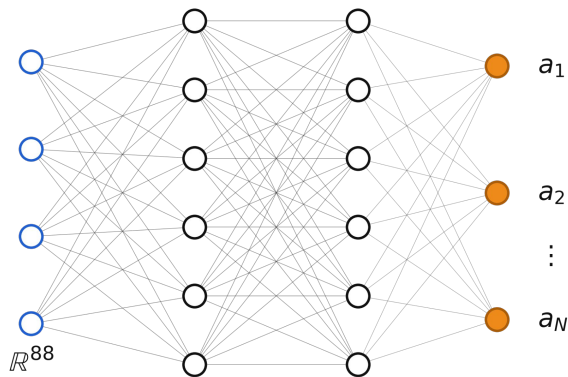
Truncated CVaR: train only the tail

- Train only the k tail quantiles (in $[0, \alpha]$); their mean is CVaR_α .
- **Re-placing**: we change which quantiles are learned (tail), not just their weights
- Statistically indistinguishable results from full CVaR — cheaper/faster, not better.



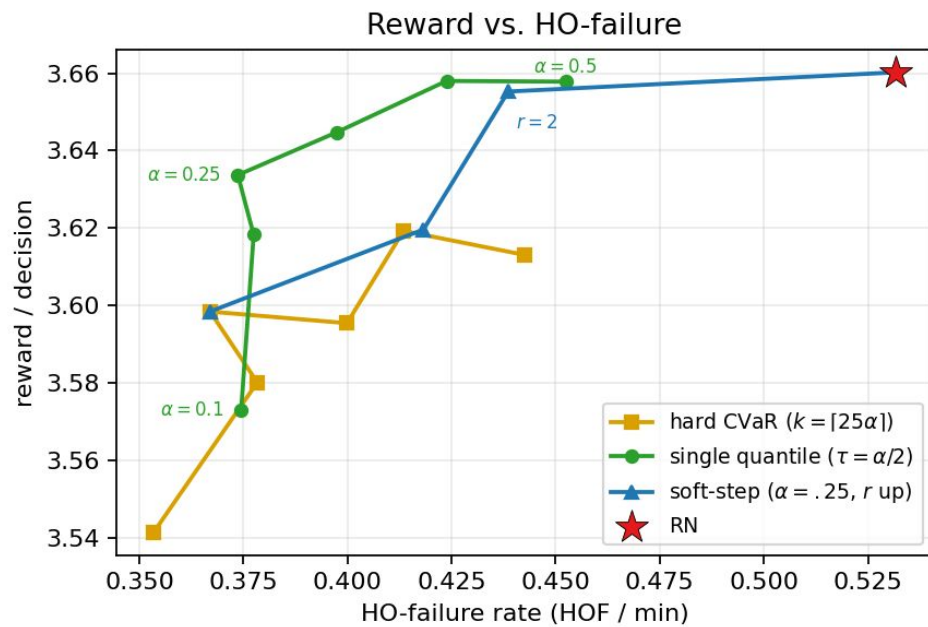
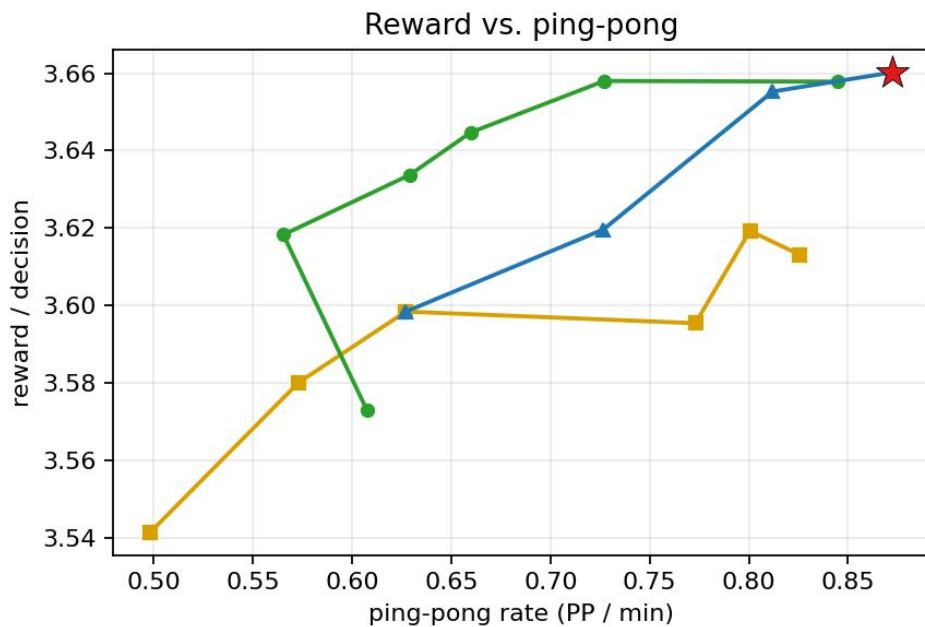
RA-1q: a single dedicated quantile

- Push it to the limit: only one quantile at $\tau = \alpha/2$, trained and acted on.
- “Learn what you act on” — the acted quantile must also be the trained one (learning the full dist but acting on one ties reward, leaves HOF higher).
- DQN's output size & inference cost, trains fastest, and the lowest failures of any learned agent (HOF 0.399, -40% vs DQN).



The Risk–Reliability Frontier

Risk frontier (efficient = upper-left)



6. Experimental Setup

The testbed

- Multi-sector 5G deployment: $7 \text{ BS} \times 3 \text{ sectors} = 21$.
- 1,000 UE trajectories, 300 s each, channel gains every 10 ms.
- Simulator: SINR $\rightarrow$ MCS $\rightarrow$ spectral efficiency; sustained outage $\rightarrow$ RLF.
- Calibrated to the reference LTM — reproduces its published KPIs.
- The agent controls only the execution decision; preparation + physics stay reference LTM.

The agents we compare

References: LTM (hand-tuned, our simulator), CMAB (KernelUCB bandit)

Learned — shared 256×256 trunk, same state/action/reward; differ only in the head & risk rule:

- DQN — scalar value
- RN — risk-neutral QR-DQN ($N=25$, mean)
- Soft-step — spectral reweighting ($\alpha=0.25$, $r=2$)
- RA-1q — single dedicated quantile ($\alpha=0.25$)

Frozen weights, $\epsilon=0$, train and test over all 1,000 UEs, mean $\pm$ std over 5 seeds.

What we measure

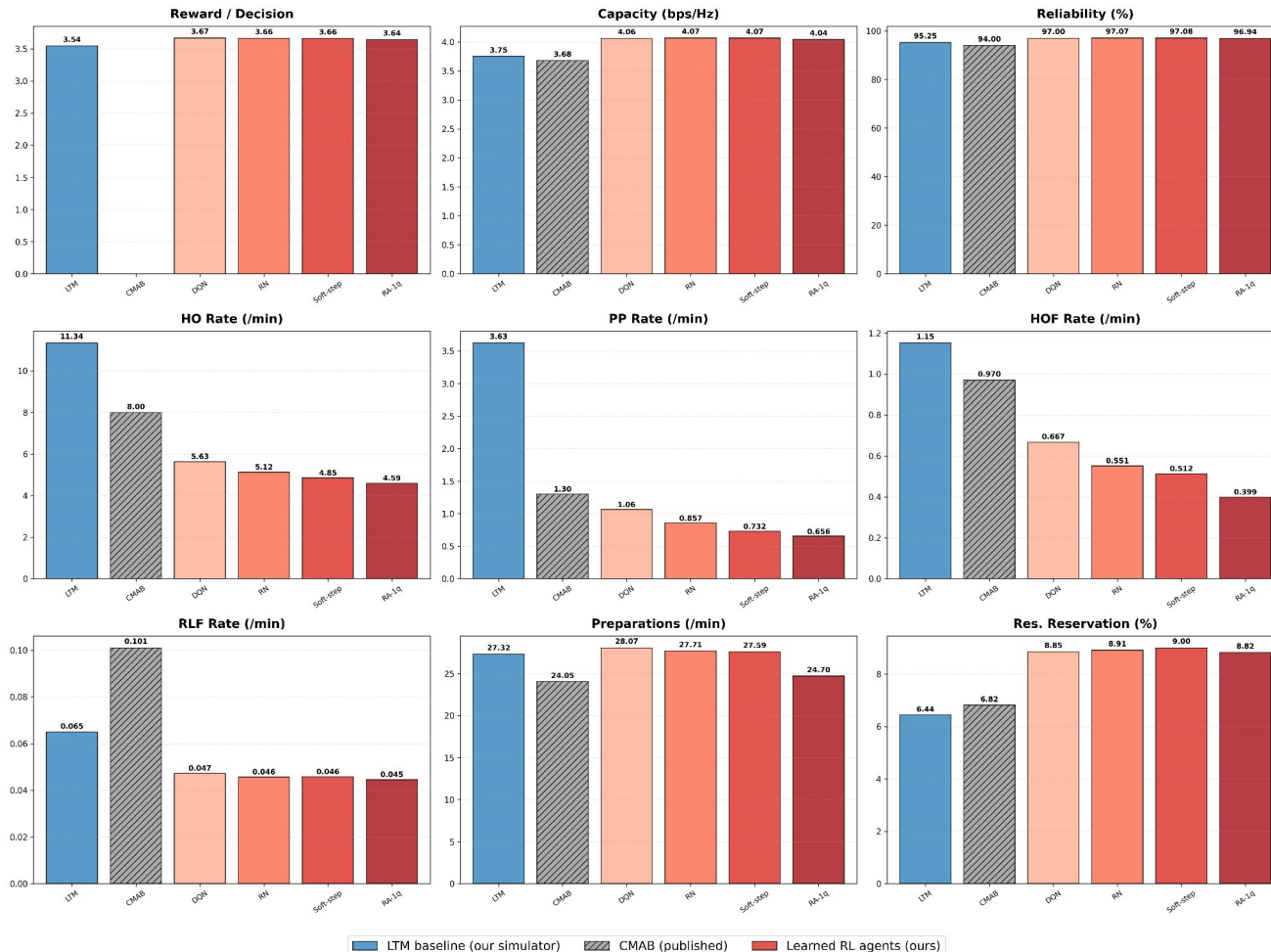
Primary: reward per decision (recall: event-driven → normalize by # decisions)

8 KPIs:

- ↑ Capacity (bps/Hz), Reliability (%)
- ↓ HO, PP, HOF, RLF (/min)
- ↓ Preparations (/min), Resource Reservation (%)

7. Results

LTM-HO Comparative Analysis: RL Agents vs. State-of-the-Art Baselines



Results

- **Signaling frugality** — the learned agents issue fewer handovers (4.6–5.6/min) than LTM (11.3) and CMAB (8.0). Reliability gains without more control-plane churn — not a signaling-for-reliability trade.
- **The one trade-off** — resource reservation: learned agents hold more cells ready (~8.8% vs 6.4% LTM). The price paid for lower failures.

Results table

TABLE V

FINAL PERFORMANCE ON THE LTM HANDOVER TASK. FOR THE LEARNED AGENTS, THE REWARD PER DECISION IS THE CROSS-SEED MEAN $\pm$ STANDARD DEVIATION OVER 2,000 EPISODES AND FIVE SEEDS (STANDARD DEVIATION ACROSS SEEDS, NOT A CONFIDENCE INTERVAL; EACH SEED IS EVALUATED ON ALL 1,000 TRAJECTORIES). **LTM**: OUR HAND-TUNED BASELINE. **CMAB***: THE CONTEXTUAL MULTI-ARMED BANDIT OF [1]. **DQN**: SCALAR BASELINE. **RN**: RISK-NEUTRAL QR-DQN ($N=25$). **SOFT-STEP**: SPECTRAL REWEIGHTING ($\alpha=0.25$, $r=2$). **RA-1q**: SINGLE DEDICATED QUANTILE ($\alpha=0.25$). THE LTM REWARD PER DECISION IS MEASURED IN OUR SIMULATOR; CMAB REPORTS NO REWARD. BOLD MARKS THE BEST VALUE IN EACH ROW ACROSS ALL AGENTS; $\uparrow/\downarrow$ GIVE THE DESIRABLE DIRECTION.

KPI	LTM	CMAB*	DQN	RN	Soft-step	RA-1q
Reward/decision $\uparrow$	3.544	—	3.667$\pm$0.006	3.663 $\pm$ 0.011	3.659 $\pm$ 0.007	3.641 $\pm$ 0.008
Capacity (bps/Hz) $\uparrow$	3.753	3.68	4.062	4.068	4.067	4.044
Reliability (%) $\uparrow$	95.25	94.0	97.00	97.07	97.08	96.94
HO/min $\downarrow$	11.34	8.00	5.63	5.12	4.85	4.59
PP/min $\downarrow$	3.625	1.30	1.065	0.857	0.732	0.656
HOF/min $\downarrow$	1.151	0.97	0.667	0.551	0.512	0.399
RLF/min $\downarrow$	0.065	0.101	0.047	0.046	0.046	0.045
Prep/min $\downarrow$	27.3	24.1	28.1	27.7	27.6	24.7
Res. reserv. (%) $\downarrow$	6.44	6.82	8.85	8.91	9.00	8.82

*CMAB figures are reproduced from [1] and were obtained in that work's simulator; our LTM baseline reproduces that work's published LTM KPIs to within $\approx 5\%$ on the radio and mobility metrics, so the CMAB column is indicative rather than strictly comparable.

8. Conclusions

One well-placed quantile is enough

- **Distribution helps — for free.** Modelling the full return (RN), still acting on the mean, lowers handovers, ping-pongs and failures at equal reward vs scalar DQN.
- **Risk-awareness is a controllable dial.** Acting on the CVaR tail trades a little reward for fewer failures — monotone, knee at $\alpha = 0.25$, no interior optimum.
- **One quantile is enough.** RA-1q — trained and acting on a single tail quantile ($\tau = \alpha/2$) — gets the lowest failures of any learned agent at plain-DQN cost.
"Learn what you act on."
- **Beats prior art. vs hand-tuned LTM:** better on every user-facing & safety KPI, with fewer handovers; vs CMAB: better on quality.

9. Limitations and Future Work

Limitations

- No held-out split — train and eval on the same fixed 1,000 trajectories. Measures how well we exploit this environment, not generalization to unseen geometry.
- Single deployment / channel dataset; reward coefficients inherited from prior work, not re-tuned.

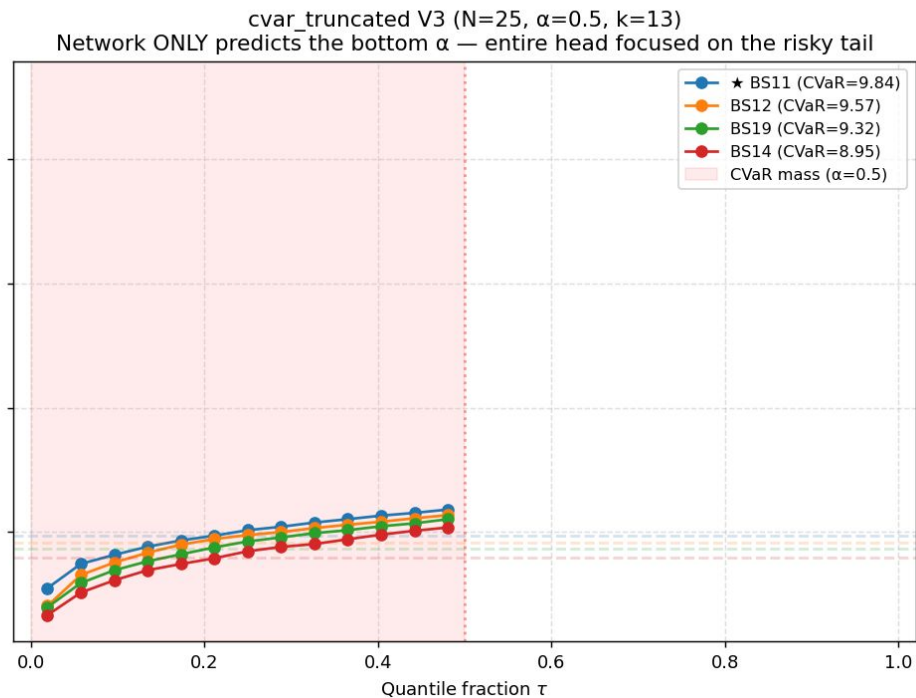
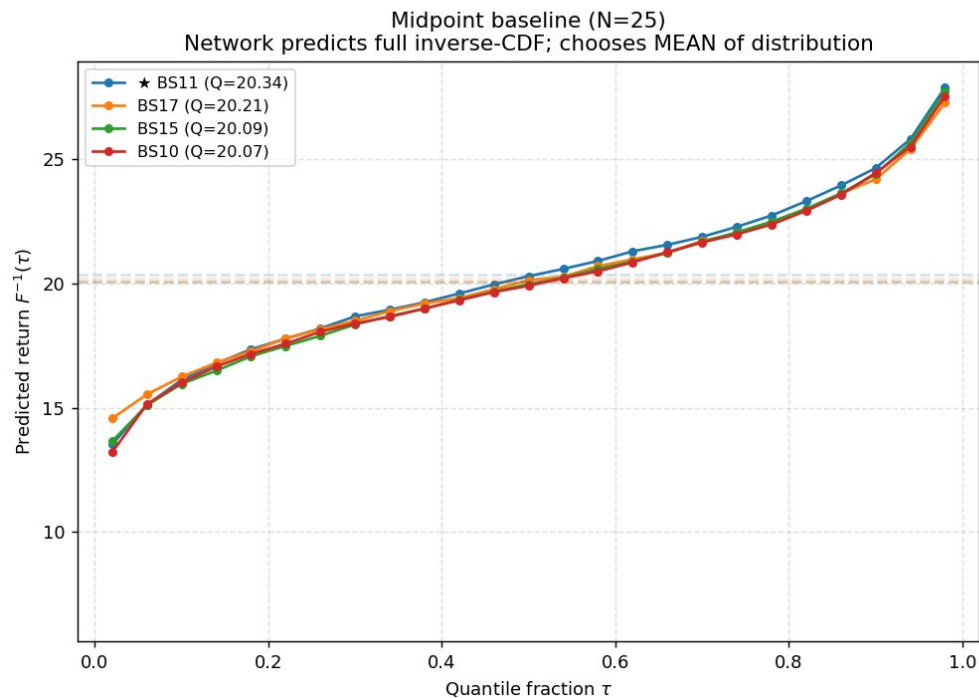
Future Work

- Learn the quantile fractions themselves (IQN / FQF) — would subsume the placement study.
- Bound failures directly via constrained / multi-objective RL, instead of through the scalar reward.
- Extend the learned policy to preparation, not just execution, and to decide after a failure takes place.
- Price reservation into the reward to close the one gap where we trail.
- Confirm the findings transfer to standard benchmarks (Atari2600).

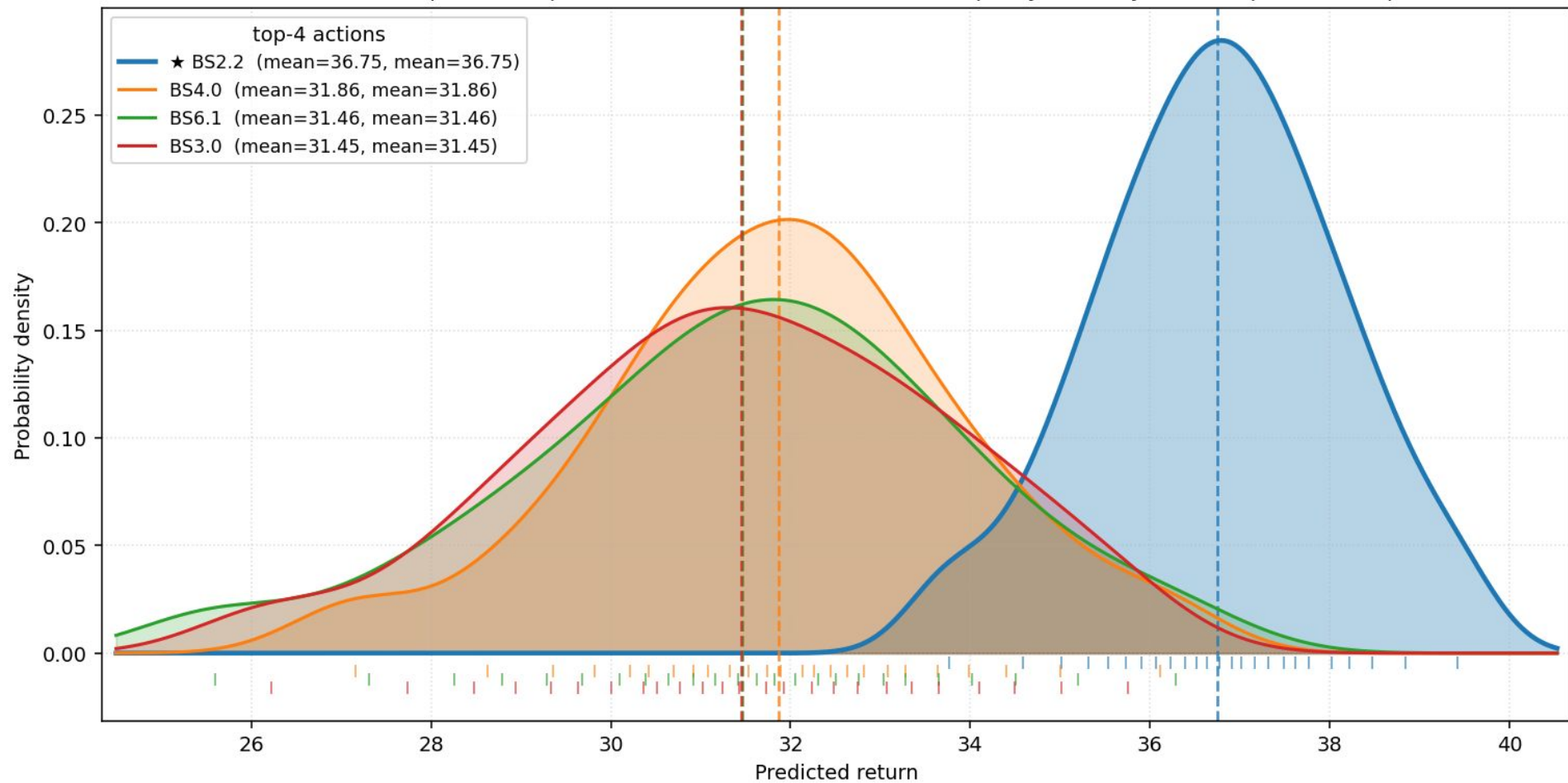
References

[1] A. Y. Moreno-Locubiche, J. Vidal, O. Muñoz-Medina, and M. Cabrera-Bean, “Contextual bandits and reconfigurable intelligent surfaces for predictive LTM handover decisions,” 2025, arXiv:2512.08556

Return-distribution comparison (UE 500, step 100) — midpoint vs cvar_truncated V3

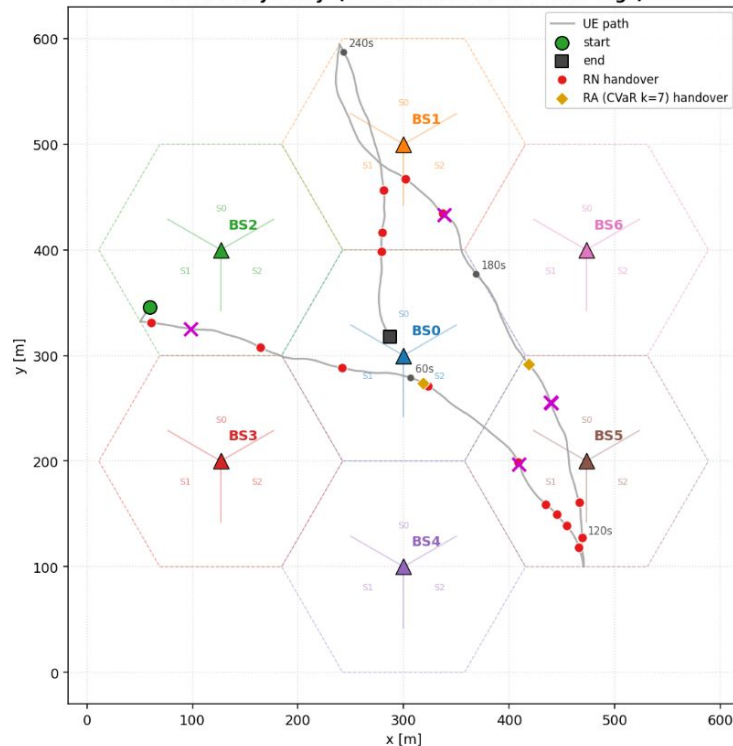


Return-distribution density per action — RN (risk-neutral QR-DQN)
(KDE over the 25 predicted quantiles; dashed line = the mean the policy ranks by; ticks = quantile samples)



RN vs. RA (CVaR k=7) — same UE 250, same trajectory, different handover decisions

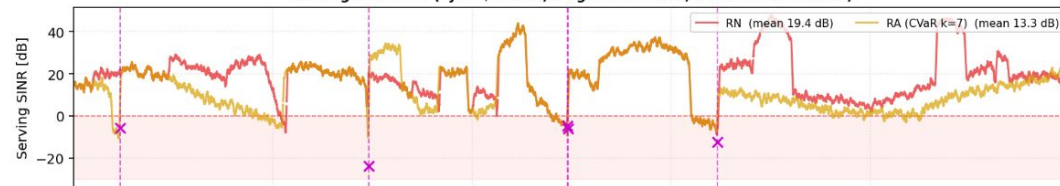
UE 250 trajectory (handover locations + time tags)



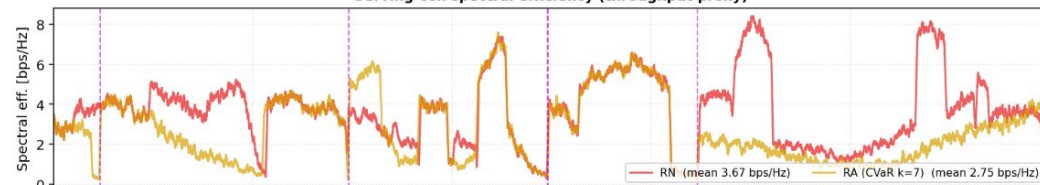
Episode scoreboard (green = better)

KPI (per UE)	RN	RA
Handovers / ep	17	2
Ping-pongs	0	0
HO failures (HOF)	0	0
Radio-link fails (RLF)	1	4
Mean SINR [dB]	19.43	13.28
Mean SE [bps/Hz]	3.67	2.75
Avg prepared cells	3.01	2.61

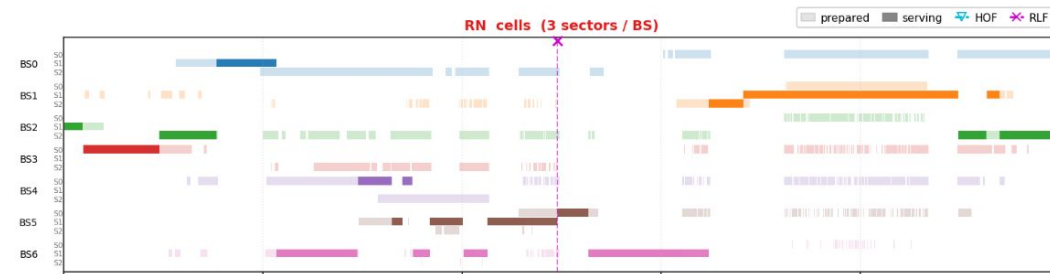
Serving-cell SINR (cyan ∇ = HOF, magenta $\times$ = RLF, shaded = OOS zone)



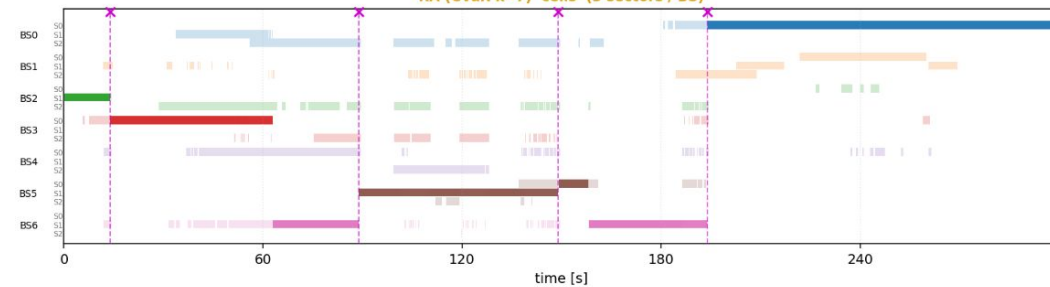
Serving-cell spectral efficiency (throughput proxy)



RN cells (3 sectors / BS)

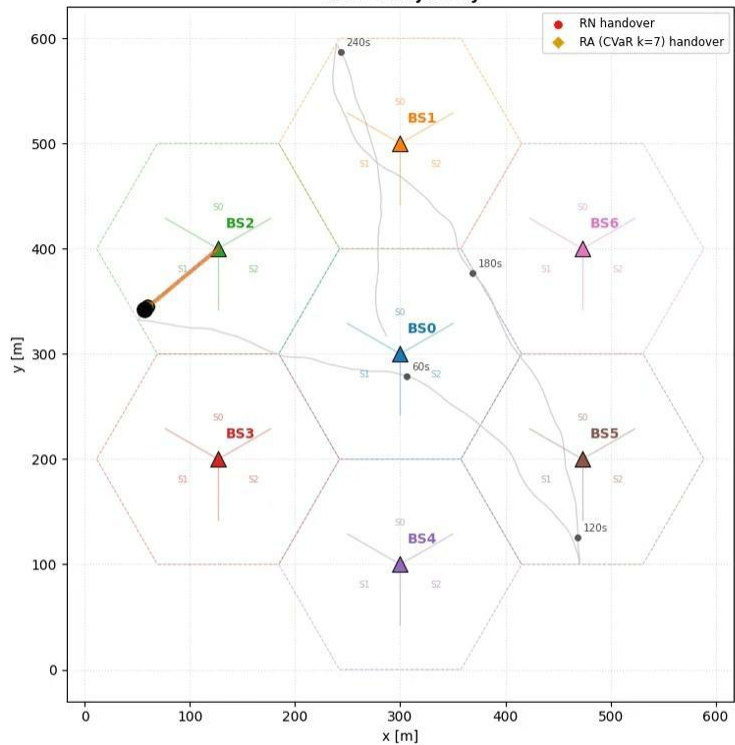


RA (CVaR k=7) cells (3 sectors / BS)



RN vs. RA (CVaR k=7) — UE 250, same trajectory

UE 250 trajectory



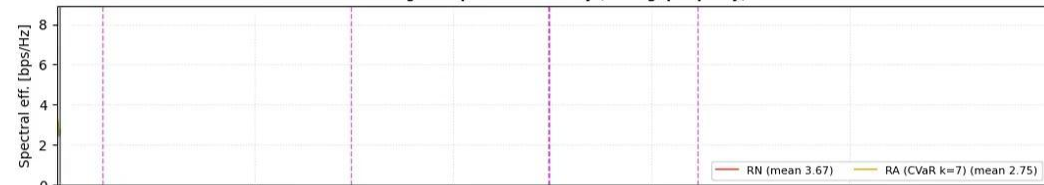
Episode scoreboard (green = better)

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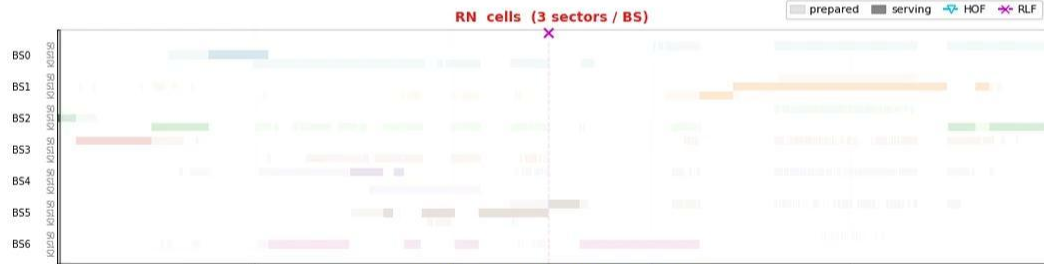
Serving-cell SINR (cyan ▽ = HOF, magenta X = RLF, shaded = OOS zone)



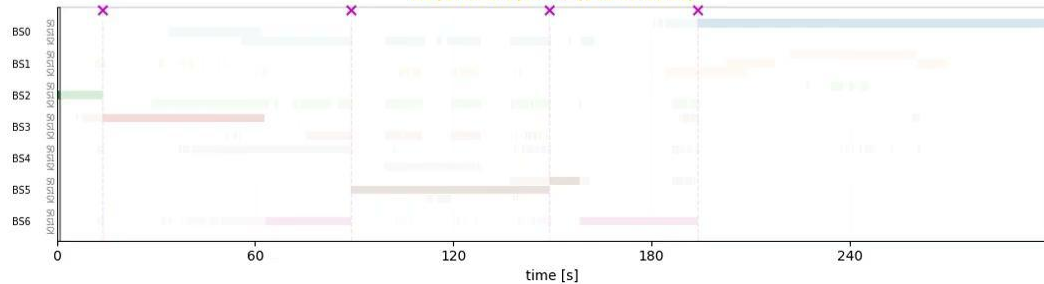
Serving-cell spectral efficiency (throughput proxy)



RN cells (3 sectors / BS)



RA (CVaR k=7) cells (3 sectors / BS)



Risk-Aware Distributional RL for 5G-Advanced Handover Decisions

Gerard Grau

Advisors: Marga Cabrera Bean, Josep Vidal